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SE Foundations - Sprint1 Report

*******Notes and LLM source for Sprint1 will be at the bottom of this document*******

I

Feature: 'Choose the board size.'
ID: 1
User Story: As a player, I want to choose a board size so that I can play on my preferred difficulty.
Priority: High
Estimated Effort(hours): 2-3 hours

Feature: 'Choose the board type.'
ID: 2
User Story: As a player, I want to choose a board type so that I can play different layouts and shapes, adding variety to the experience.
Priority: High
Estimated Effort(hours): 2-4 hours

Feature: 'Start a new game of the chosen board size and type.'
ID: 3
User Story: As a player, I want to start a new game at the chosen configuration so that I can begin playing Peg Solitaire.
Priority: High
Estimated Effort(hours): 1-2 hours

Feature: 'Making a move.'
ID: 4
User Story: As a player, I want to make a valid move on the board so that we can make progress in the game.
Priority: High
Estimated Effort(hours): 2-4 hours

Feature: ‘Determining if the game is over.’
ID: 5
User Story: As a player, I want the program to detect when there are no valid moves left, so that we can conclude the game is over.
Priority: High
Estimated Effort(hours): 2-4 Hours

II

User Story ID/Name: 1. Choose a board size

- AC ID/Description: 1.1 Valid board size selection is accepted
    - Status: toDo
    - Given that the game is on the main screen before a new game starts, with options
    - When the player enters/selects a valid board size value
    - Then the system stores that value as the “selected board size” for the next new game

- AC ID/Description: 1.2 Invalid board size selection is rejected
    - Status: toDo
    - Given that the game is on the main screen before a new game starts
    - When the player enters/selects a board size value that is not a valid
    - Then the system does not update the “selected board size” provides clear feedback on invalid input

- AC ID/Description: 1.3 Changing the board size does affect the current game
    - Status: toDo
    - Given that a game is currently in progress
    - When the player changes the board size input control
    - Then the current board state on screen does not change

- AC ID/Description: 1.4 Default board size
    - Status: toDo
    - Given that the program has started
    - When the player has not chosen a board size and starts the game
    - Then the game starts a new game with a board size of the default value

User Story ID/Name: 2. Choose a board type

- AC ID/Description: 2.1 Valid board type selection is accepted
    - Status: toDo
    - Given that the game is on the main screen before a new game starts, with options
    - When the player enters/selects a valid board type value
    - Then the system stores that value as the “selected board type” for the next new game

- AC ID/Description: 2.2 Invalid board type selection is rejected
    - Status: toDo
    - Given that the game is on the main screen before a new game starts
    - When the player enters/selects a board type value that is not a valid
    - Then the system does not update the “selected board type” provides clear feedback on invalid input

- AC ID/Description: 2.3 Changing the board type does affect the current game
    - Status: toDo
    - Given that a game is currently in progress
    - When the player changes the board type input

        - Then the current board game or state on screen does not change
    - AC ID/Description: 2.4 Default board type exists at startup
        - Status: toDo
        - Given that the program has started
        - When the player has not chosen a board type and starts the game
        - Then the game starts a new game with a board type of a default value

User Story ID/Name: 3. Starting a new game
- AC ID/Description: 3.1 New game initialization
    - Status: toDo
    - Given player has chosen board size and type
    - When the player selects 'start game'
    - Then the program should display the board of the selected type and size

- AC ID/Description: 3.2 New game with unselected type/size
    - Status: toDo
    - Given that the player has not selected either type or size
    - When the player initiates a new game
    - Then the program should create a game displaying a board of default type or size, or both

- AC ID/Description: 3.3 Starting a new game will reset the entire progress/state of the previous session
    - Status: toDo
    - Given that a game is ongoing
    - When the player chooses to start a new game
    - Then discard the current board, moves, and state of the game to initiate a new game with a fresh state.

<u>User Story ID/Name:</u> 4. Making a move in the game

- <u>AC ID/Description</u>: 4.1 Valid moves
    - <u>Status</u>: toDo
    - Given that the game is ongoing AND there is a peg A that can jump over adjacent peg B into an empty valid position
    - When the player selects or chooses said move
    - Then the game has peg A occupy a new position, remove peg B, and positions A and B are empty

- <u>AC ID/Description</u>: 4.2 One move per player
    - <u>Status</u>: toDo
    - Given that the game is ongoing
    - When the player enters a valid move
    - Then the game executes only that move AND prompts the next player to initiate a move

- <u>AC ID/Description</u>: 4.3 Restriced moves
    - <u>Status</u>: toDo
    - Given that a game is ongoing
    - When the player enters a diagonal, and the board is not the correct type
    - Then the game rejects the move and waits for a valid move

- <u>AC ID/Description</u>: 4.4 Invalid move will not update the board or game
    - <u>Status</u>: toDo
    - Given that the game is ongoing
    - When the player selects a restricted move
    - Then the game state is unchanged, and board stays the same with the pegs in the same positions

User Story ID/Name: 5. Determining if a game is over

- AC ID/Description: 5.1 No more valid moves
  - Status: toDo
  - Given that the game is ongoing
  - When there are no more possible moves to be made on the board
  - Then the program determines it has completed and displays the ‘game over’ message

- AC ID/Description: 5.2 One peg left
  - Status: toDo
  - Given that the game is ongoing
  - When there is only one peg left on the board
  - Then the program determines the session is over with the puzzle solved! Congrats message displayed!

- AC ID/Description: 5.3 Menu Reset
  - Status: toDo
  - Given that a game is there is no more moves to be made
  - When the program determines a complete game
  - Then redirect to the main menu, displaying the homepage with board type/size options for a new game

- AC ID/Description: 5.4 Completion detection
  - Status: toDo
  - Given that a valid move was just made
  - When the state of the game and the board are updated
  - Then the program always checks whether valid moves are left or if there is more than one peg left AND if so, update status of the game

Notes: I used an LLM to help draft user stories and acceptance criteria for the 'Choose a board size' and 'Determining if a game is over' features. I made sure the response of the LLM made sense and successfully met the task. I adjusted several criteria of the output and added significant changes and adjustments. The final version of this Sprint1 report reflects my understanding of the requirements.

Here is a link to my LLM collaboration: [ChatGPT LLM link]

I'm open to refining how I use LLM tools in future sprints to make sure I'm balancing understanding, but also efficiency. I am not sure if I should do more with LLMs; maybe I should have used it for all 5 features, and asked for even more clarity for Acceptance Criteria.

'Priority' for all these features is high and needs to be implemented for a complete program. 'Estimated hours' are loose estimates, as I have little experience with this, but I understand the importance of the category.

I also didn't use the tables given in Sprint1 as this was a simpler format. If you need me to convert, I can do so. Please let me know.

Thanks!